1.1 What is D3.js?

D3.js was created to fill a pressing need for web-accessible, sophisticated data visualization. Because of the library's robust design, it does more than make charts. And that's a good thing, because *data visualization* no longer refers to pie charts and line graphs. It now means maps and interactive diagrams and other tools and content integrated into news stories, data dashboards, reports, and everything else you see on the web.

D3.js's creator, Mike Bostock, helped develop an earlier data visualization library, Protovis, and also developed Polymaps, a JavaScript library that provides vector- and tile-mapping capability in a lightweight form. These earlier endeavors would inform the creation of D3.js, which focuses on modern standards and modern browsers. As Bostock describes it, "This avoids proprietary representation and affords extraordinary flexibility, exposing the full capabilities of web standards such as CSS3, HTML5 and SVG" (http://d3js.org/). This is the radical nature of D3.js. Although it won't run on Internet Explorer 6, the widespread adoption of standards on modern browsers has finally allowed web developers to deliver dynamic and interactive content seamlessly in the browser.

Until recently, you couldn't build high-performance, rich internet applications in the browser unless you built them in Flash or as a Java applet. Flash and Java are still around on the internet, and especially for internal web apps, for this reason. D3.js provides the same performance, but integrated into web standards and the Document Object Model (DOM) at the core of HTML. D3 provides developers with the ability to create rich interactive and animated content based on data and tie that content to existing web page elements. It gives you the tools to create high-performance data dashboards and sophisticated data visualization, and to dynamically update traditional web content.

But D3 isn't easy for people to pick up, because they often expect it to be a simple charting library. A case in point is the pie chart layout, which you'll see in chapter 5. D3 doesn't have one single function to create a pie chart. Rather, it has a function that processes your dataset with the necessary angles so that, if you pass the dataset to D3's `arc` function, you get the drawing code necessary to represent those angles. And you need to use yet another function to create the paths necessary for that code. It's a much longer process than using dedicated charting libraries, but the D3 process is also its strength. Although other charting libraries conveniently allow you to make line graphs and pie charts, they quickly break down when you want to make something more than that. Not D3, which allows you to build whatever data-driven graphics and interactivity you can imagine, and that's why D3 is behind much of the most innovative and exciting information visualization on the web today.

1.2 How D3 works

Let's take a look at the principles of data visualization, as well as how D3 works in general. In figure 1.1 you see a rough map of how you might start with data and use D3 to process and represent that data, as well as add interactivity and optimize the data visualization you've created. In this chapter we'll start by establishing the principles of how D3 selections and data-binding work and learning how D3 interacts with SVG and